View Reviews

Paper ID
187

Paper Title
Comparative Deep Neural Study For Stage-Wise Alzheimer's Disease Detection Using Brain MRI

Reviewer #1

Questions

1. Review Comments

The dataset balancing process increases Mild, Moderate, and Very Mild classes far beyond their real counts, but the manuscript does not describe the oversampling method, making the extreme upscaling (e.g., 488 → 8000) potentially unreliable and prone to data duplication.

Table 5 shows unusually high accuracy values (0.95–0.98) for most models despite major class imbalance and simple preprocessing, suggesting possible data leakage or insufficient separation between training and test sets.

The test-sample counts listed in Table 5 are inconsistent; DenseNet201 is evaluated on 160 samples while other models use 15,600 samples, preventing fair comparison.

The description of training configuration (Table 4) is incomplete and inconsistent; for example, different loss functions are listed for similar models but the manuscript does not justify why SparseCategoricalCrossentropy is used for some models and CategoricalCrossentropy for others.

The methodology does not specify how MRI slices were selected, whether each patient contributes multiple slices, or whether patient-wise splitting was used, which is critical to prevent identity leakage.

The evaluation metrics do not include ROC-AUC, class-wise performance, or confusion matrices for each model, even though the problem is a four-class medical diagnosis task where class-specific errors matter significantly.

Figures showing accuracy and loss curves (Figs. 4 and 5) display perfect alignment of training and validation curves, which is uncommon and may indicate data mixing or insufficient validation independence.

The conclusion incorrectly states that EfficientNet-B1 outperformed simpler models, even though Table 5 shows the opposite; this contradiction indicates issues in result interpretation.

2. Recommendation: Your overall rating
Major Revision

Reviewer #2

Questions

1. Review Comments
The related-work section summarizes many external studies but does not identify a clear research gap, leaving it unclear what new scientific contribution this comparative analysis provides.

The resizing of images to 128×128 or 224×224 ignores MRI spatial structure; the paper does not mention skull stripping, intensity normalization, or slice-position standardization, which are normally required for MRI-based AD classification.

The models evaluated (ResNet50, DenseNet201, VGG16, etc.) are standard 2D CNNs, yet MRI is a 3D volumetric modality; reducing it to 2D slices without explaining slice selection severely limits diagnostic relevance.

No explanation is provided for why InceptionV3 and EfficientNet-B1 perform poorly (0.29–0.75 recall), and the manuscript does not analyze whether hyperparameters, data preprocessing, or image resolution contributed to the drop.

The class names in Table 5 include duplicate precision/recall/F1 columns, suggesting formatting issues or incorrect metric reporting.

The dataset description does not clarify whether the ADNI subset was pre-segmented or raw; no mention is made of scanner variability, intensity non-uniformity, slice thickness, or preprocessing pipelines commonly used in AD research.

No statistical tests are provided to confirm differences between models, despite extremely close values (0.98 vs 0.98), making it impossible to assess whether improvements are meaningful.

The workflow diagram (Fig. 2) oversimplifies the classification process and omits key steps such as data normalization ranges, augmentation parameters, class-weighting strategies, and validation procedures.

2. Recommendation: Your overall rating

Minor Revision

Reviewer #3

Questions

1. Review Comments

Too generic — revise it to reflect novelty and experimental outcome precisely. Lacks specificity revise to highlight the innovative aspects and clearly articulate the experimental results

A better contextual background with real-time application examples would improve engagement.

Clearly define all variables and ensure each equation is numbered and referred to appropriately in the text.

Consider adding visual results (e.g., segmentation maps, classification heatmaps) to substantiate quantitative results.

There is minimal comparison with existing methods. Discuss how your work outperforms or complements prior research.

Some tables are dense and hard to interpret. Improve formatting and use graphical plots where possible.

It should include future work and limitations instead of merely restating findings.

The number of references should be increased to a minimum of 15 citations within the paper.

Provide proper reference, Don’t use et al in the author position.

Figure number 1(a), 1(b), 1(c), 1(d) is in-text description/callout missing in the paper.

All Figures quality should be improve in the paper.

Equation Number 1-5 callout is missing in the paper

2. Recommendation: Your overall rating

Major Revision